

CIS 6270

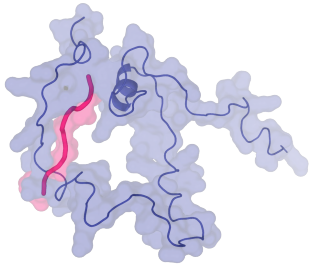
Module 4: Discrete Diffusion

Lecture 4.5: Learned Rates, Planners, and Guided Sequence Generation

Tue, Oct 20, 2026

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Should every token disappear at the same rate?

MD4 and GenMD4

Simplified and Generalized Masked Diffusion for Discrete Data

Jiaxin Shi*, **Kehang Han***, **Zhe Wang**, **Arnaud Doucet**, **Michalis K. Titsias**
Google DeepMind

Shi,..,Titsias. *NeurIPS* (2024)

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The generalized loss has a state-dependent weight, and we learn the schedule itself with the same bound.

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A slower masking rate keeps a token visible longer, so other positions can use it as context.

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A token-dependent schedule controls which information remains available during corruption.

It has a continuous-time bound with a weight that depends on the token

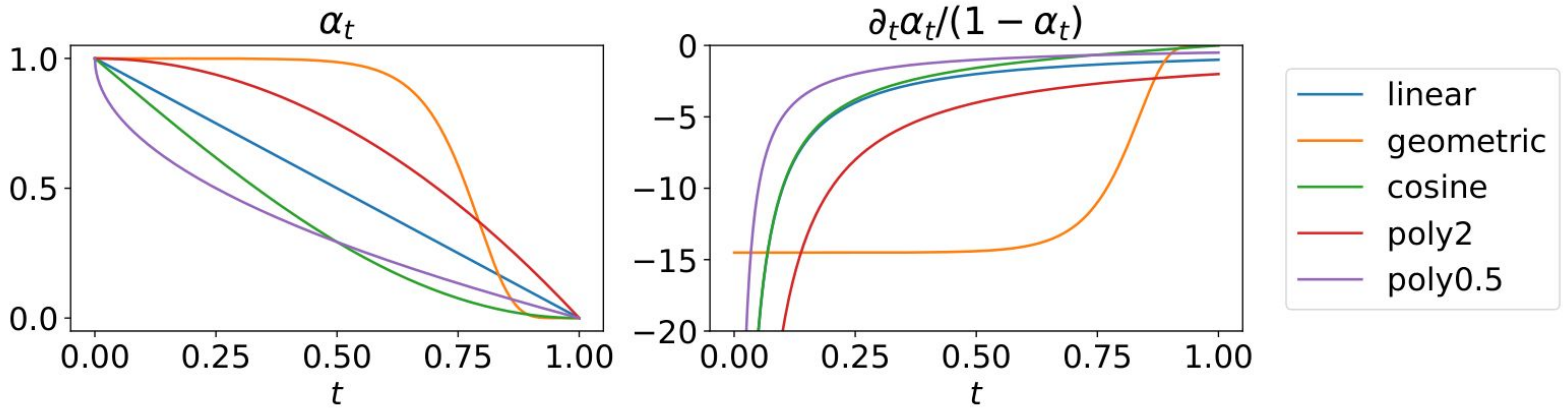


Figure 1: Masking schedules in the literature: (Left) α_t; (Right) weight of the cross-entropy loss w.r.t. t; Equations for these schedules are given in Tab. 4 in Appendix.

In the generalized model, GenMD4, we learn the per-token schedule.

Shi,..,Titsias. *NeurIPS* (2024)

Here is the Text8 result in Table 2 of the paper

Table 2: Bits Per Character (BPC) on Text8 test set. All models use standard 12-layer transformers similar to GPT-2 small [57] except Discrete Flow which uses 8×3 layers.

Method	BPC (↓)
Continuous Diffusion	
Plaid [22] (Our impl.)	≤ 1.48
BFN [26]	≤ 1.41
Any-order Autoregressive	
ARDM [48]	≤ 1.43
MAC [61]	≤ 1.40
Autoregressive	
IAF/SCF [62]	1.88
AR Argmax Flow [15]	1.39
Discrete Flow [59]	**1.23**
Transformer AR [14]	**1.23**
Discrete Diffusion	
Mult. Diffusion [15]	≤ 1.72
D3PM Uniform [14]	≤ 1.61
D3PM Absorb [14]	≤ 1.45
SEDD Absorb [32]	≤ 1.39
MD4 (Ours)	$\leq$ **1.37**
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Absorbing D3PM has a Text8 bound of 1.45 bits per character

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MD4 reaches 1.37 with the same transformer

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GenMD4, with learned per-token rates, reaches 1.34

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Choosing the rates well is worth as much as changing the model

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Shi,..,Titsias. *NeurIPS* (2024)

Could we keep several tokens plausible until the surrounding sequence makes the choice clearer?

Simplex Diffusion Models

SIMPLEX DIFFUSION MODELS

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Google DeepMind
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Google DeepMind
UCL Gatsby

Andrew Campbell
Google DeepMind

Li Kevin Wenliang
Google DeepMind

James Thornton
Google DeepMind

Arnaud Doucet
Google DeepMind

Valentin De Bortoli
Google DeepMind

Deschenaux,..,De Bortoli. *arXiv* (2026)

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This is called information collapse, because uncertainty does not carry from one step to the next.

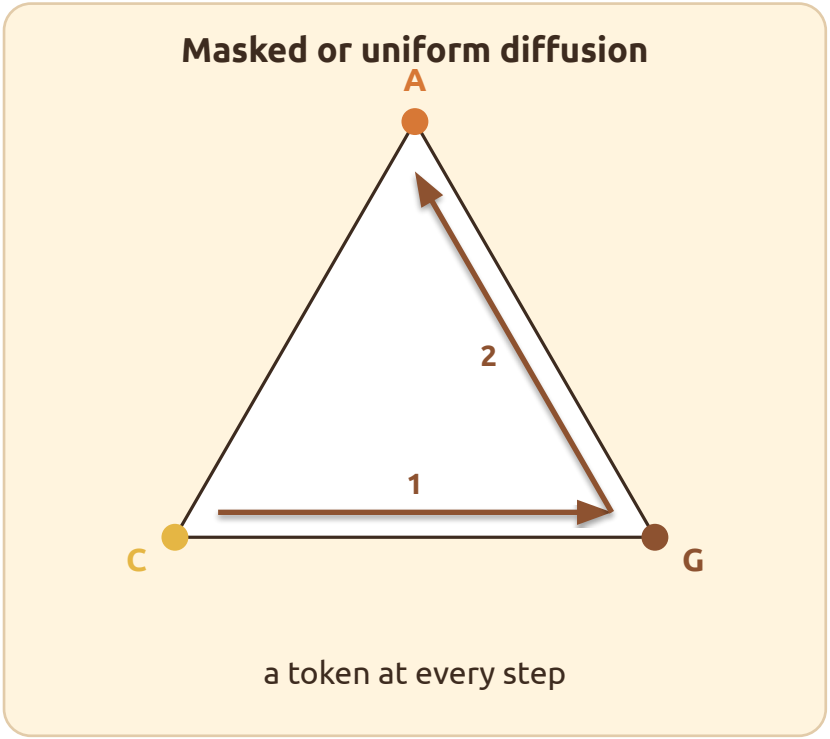
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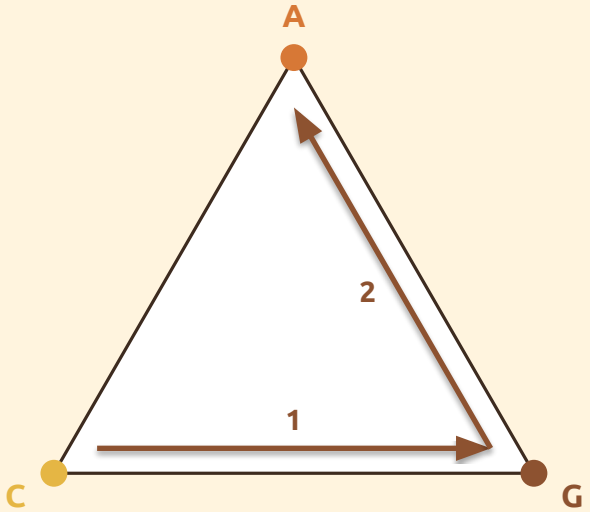
In Simplex Diffusion Models, we keep a probability vector at every position, so the state itself holds the uncertainty.

Masked and uniform diffusion move between the corners of the simplex



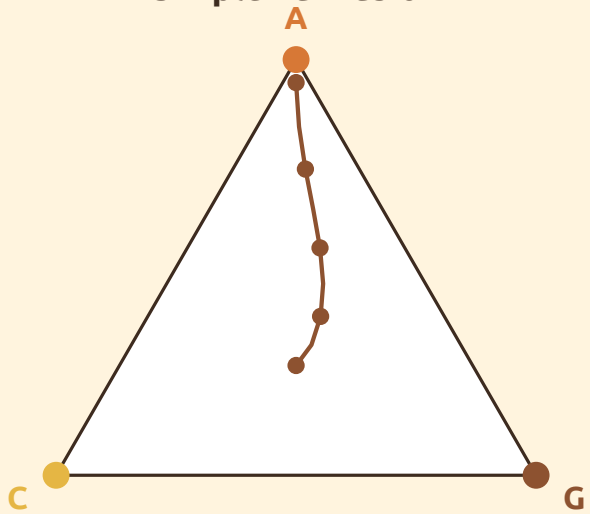
In simplex diffusion, a belief moves through its interior

Masked or uniform diffusion



a token at every step

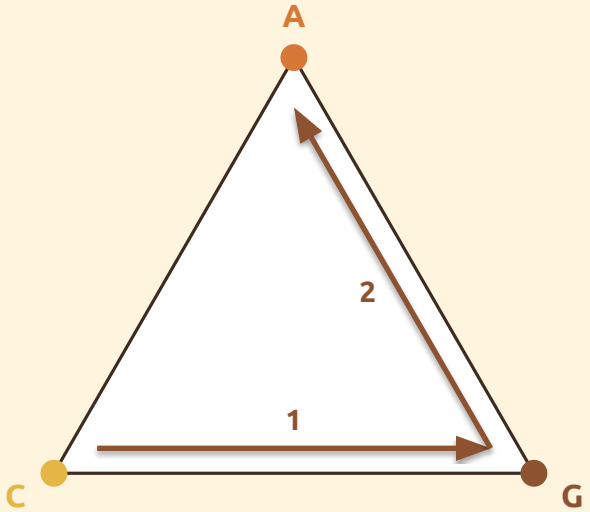
Simplex diffusion



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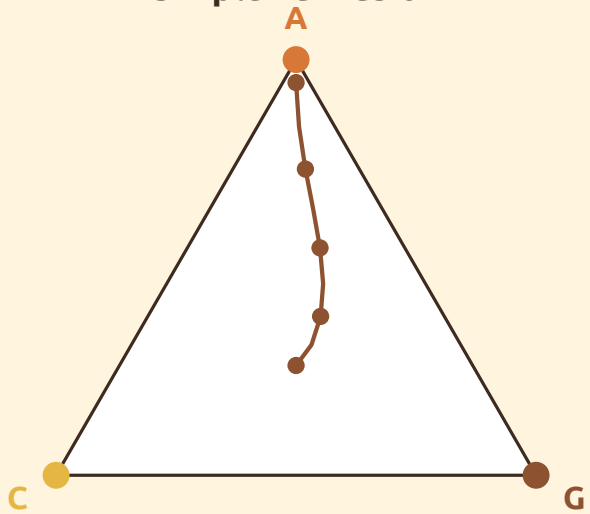
The state space changes from tokens to beliefs

Masked or uniform diffusion



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Simplex diffusion



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As the concentration goes to zero, every belief lands on a corner with the uniform-diffusion probabilities.

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A belief can keep both A and G plausible while the rest of the sequence provides more evidence.

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The forward marginal, the posterior average, and the cross-entropy are the same three pieces we used for MDLM.

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Each possible clean token proposes a reverse step, and the denoiser determines how much weight that proposal receives.

Simplex diffusion: averaging a belief update

Toy denoiser probabilities for A, C, G, T are (0.1, 0.2, 0.6, 0.1). Let B mean that the next belief puts more than half its mass on G. Given the clean base, the four probabilities of B are (0.1, 0.2, 0.8, 0.1).

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If the clean base was G, this example contributes 0.511 nats of cross-entropy.

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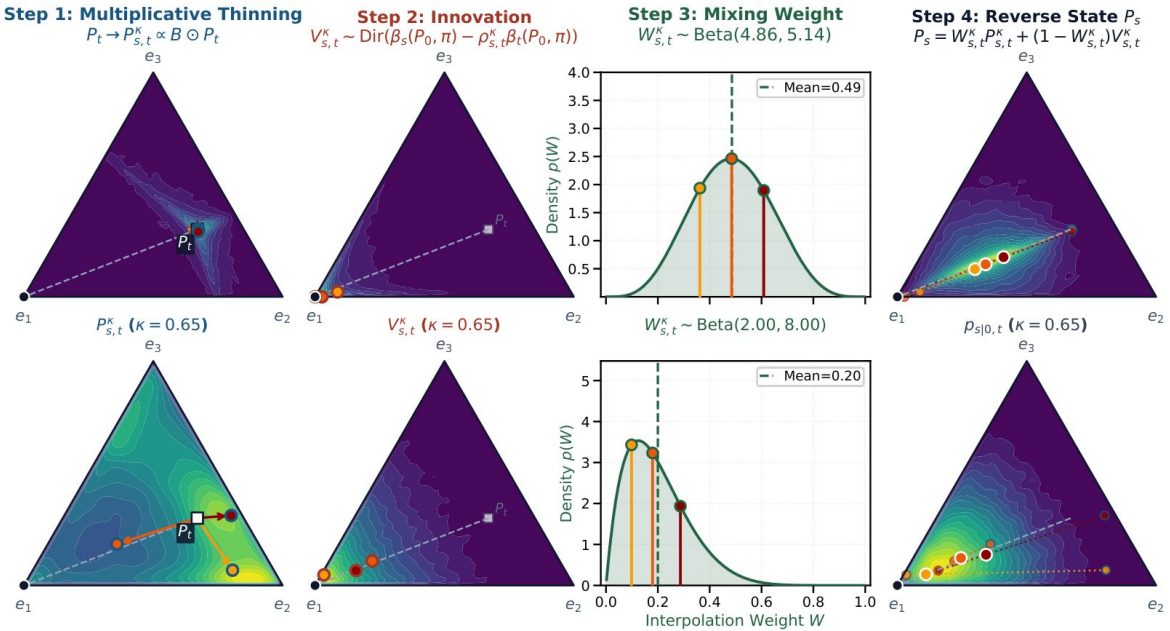
Average the four possible reverse updates with the denoiser weights. Event B gets probability 0.54.

$$x_0 = \mathrm{G}: \qquad -\log d_\theta(\mathrm{G} \mid P_t, t) = -\log(0.6) \approx 0.511$$

If the clean base was G, this example contributes 0.511 nats of cross-entropy.

The model averages over uncertainty about the clean base before it updates the noisy belief.

In one reverse step, we thin the current belief, draw an innovation, and mix the two



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Here is Figure 1 of the paper, on reasoning problems

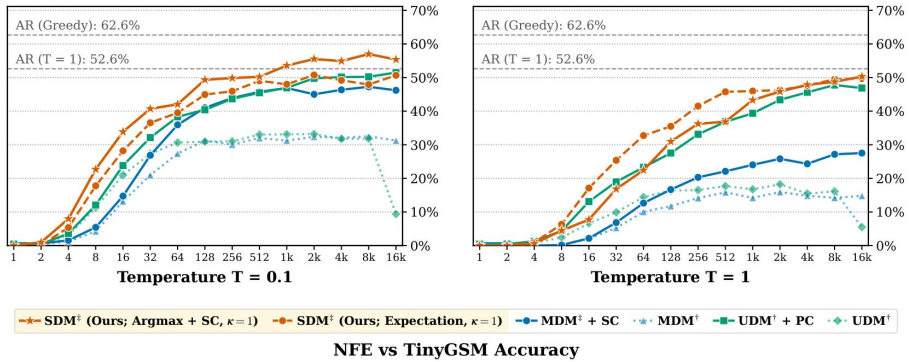


Figure 1: **SDMs outperform Discrete Diffusion on TinyGSM.** *Left:* Accuracy as a function of the number of function evaluation (NFE) at low temperature ($T = 0.1$). *Right:* Accuracy against NFEs at $T = 1$. Without Self-Conditioning (SC), SDMs with the expected embedding outperform masked diffusion with SC (MDM + SC) and uniform diffusion with predictor-corrector sampling (UDM + PC). At 512 NFE, SDMs reach 49.0% at $T = 0.1$, vs. 45.8% for MDM‡ + SC, and 45.8% at $T = 1$, vs. 36.8% for UDM† + PC. With SC, SDM (Argmax + SC) is the best diffusion variant at $T = 0.1$ from 32 NFE onwards, and solves 57% of problems with 8k NFEs. Autoregressive models with greedy decoding is best (62.6%). All SDMs use churn $\kappa = 1$ and the adaptive sampling time grid (Section J.1), and we compare with the best baseline configurations (Section 5.2). $^{\dagger}/^{\ddagger}$: trained with the uniform/adaptive time sampler. The accuracy drop for UDM at 16k is consistent with typical floating-point rounding errors, which can accumulate at very small step sizes (Karras et al., 2022).

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Accuracy on TinyGSM rises with the number of function evaluations

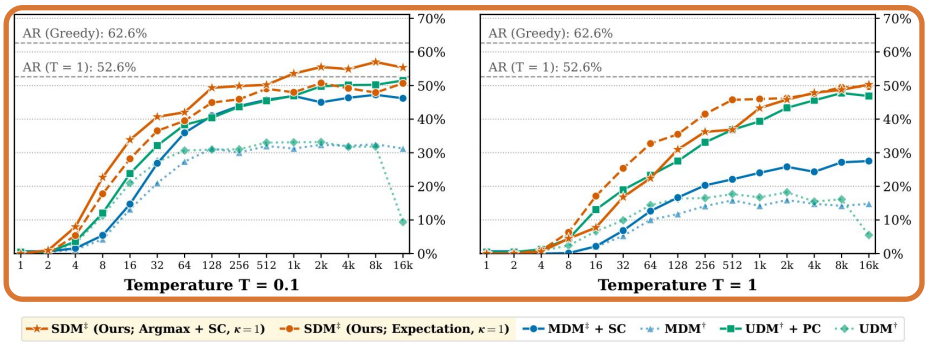


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At 512 evaluations, simplex diffusion solves 49.0% against 45.8% for masked diffusion

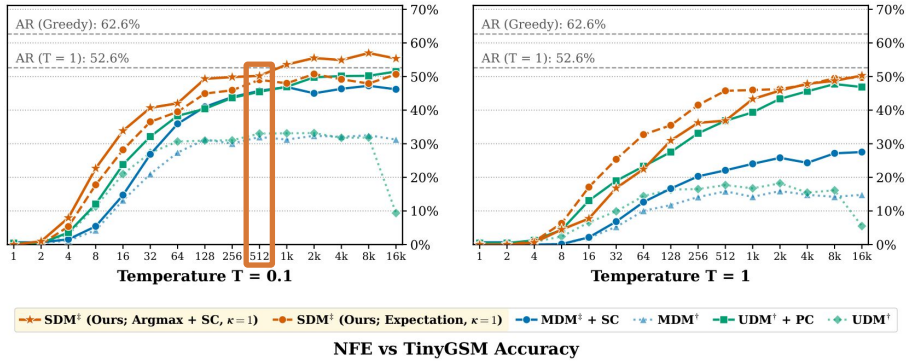


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With self-conditioning it reaches 57% at 8k evaluations

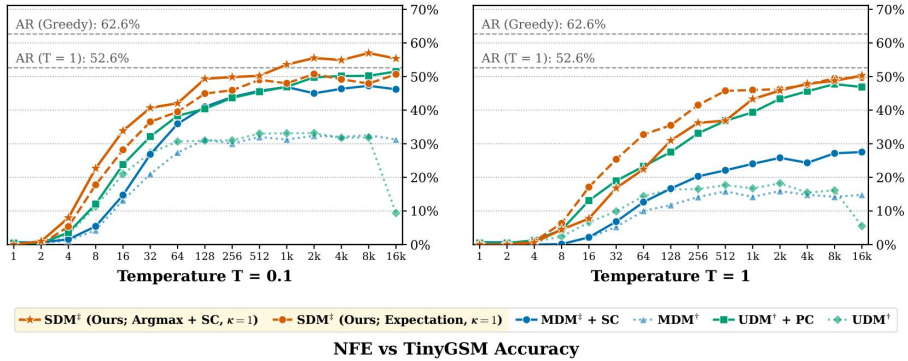


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Greedy autoregressive decoding is still ahead, at 62.6%

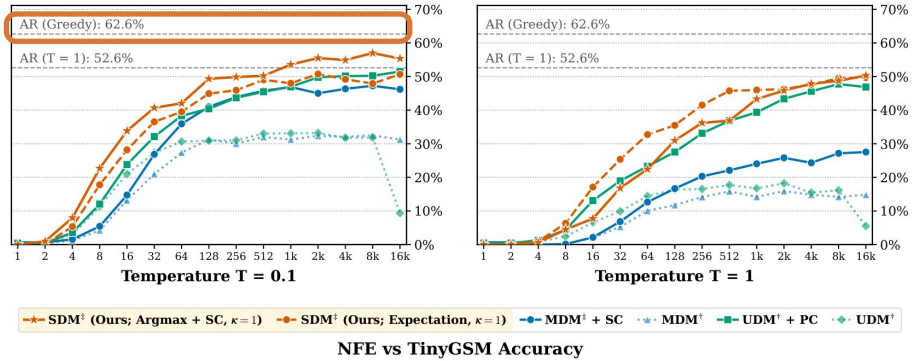


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With beliefs, we carry the denoiser's uncertainty from one step to the next

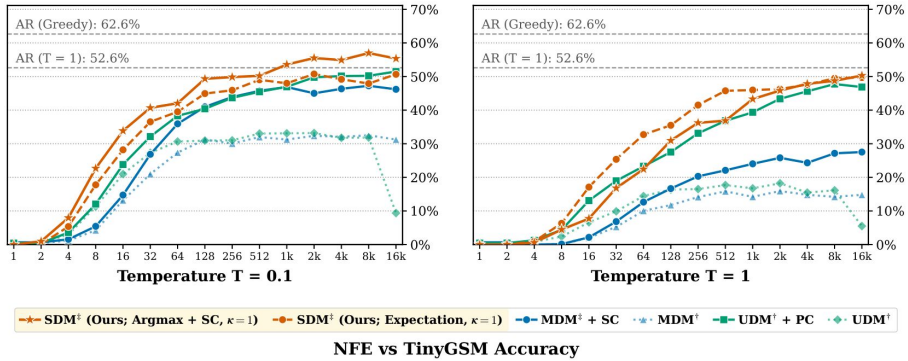


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Let’s write the whole model for DNA

```
MASK = 4                              # A, C, G, T, then the mask

def corrupt(x0, t):
    alpha = 1.0 - t                   # linear schedule
    keep = torch.rand(x0.shape, device=DEV) < alpha
    return torch.where(keep, x0, MASK)

def loss(model, x0):
    t = torch.rand(x0.shape[0], device=DEV)
    zt = corrupt(x0, t[:, None])
    logits = model(zt, t)             # [B, L, 5]
    logits[..., MASK] = -1e9          # no mass on the mask
    ce = F.cross_entropy(logits.transpose(1, 2), x0,
                         reduction='none')
    w = 1.0 / t                       # -alpha_dot/(1 - alpha)
    return (w[:, None] * ce * (zt == MASK)).sum(-1).mean()

@torch.no_grad()
def sample(model, B, L, steps=32):
    z = torch.full((B, L), MASK, device=DEV)
    ts = torch.linspace(1.0, 1e-3, steps + 1, device=DEV)
    for t, s in zip(ts[:-1], ts[1:]):
        logits = model(z, t.expand(B))
        logits[..., MASK] = -1e9
        x0 = Categorical(logits=logits).sample()
        unmask = torch.rand(z.shape, device=DEV) < (t - s) / t
        z = torch.where((z == MASK) & unmask, x0, z)
    return z
```

Our alphabet is four bases and one mask

- The mask is token 4, the extra absorbing state.

```
MASK = 4                                  # A, C, G, T, then the mask

def corrupt(x0, t):
    alpha = 1.0 - t                       # linear schedule
    keep = torch.rand(x0.shape, device=DEV) < alpha
    return torch.where(keep, x0, MASK)

def loss(model, x0):
    t = torch.rand(x0.shape[0], device=DEV)
    zt = corrupt(x0, t[:, None])
    logits = model(zt, t)                 # [B, L, 5]
    logits[..., MASK] = -1e9              # no mass on the mask
    ce = F.cross_entropy(logits.transpose(1, 2), x0,
                         reduction='none')
    w = 1.0 / t                           # -alpha_dot/(1 - alpha)
    return (w[:, None] * ce * (zt == MASK)).sum(-1).mean()

@torch.no_grad()
def sample(model, B, L, steps=32):
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    ts = torch.linspace(1.0, 1e-3, steps + 1, device=DEV)
    for t, s in zip(ts[:-1], ts[1:]):
        logits = model(z, t.expand(B))
        logits[..., MASK] = -1e9
        x0 = Categorical(logits=logits).sample()
        unmask = torch.rand(z.shape, device=DEV) < (t - s) / t
        z = torch.where((z == MASK) & unmask, x0, z)
    return z
```

To corrupt, we sample the forward marginal directly

- The mask is token 4, the extra absorbing state.
- Each position survives with probability $\alpha_t = 1 - t$, independently.

```python
MASK = 4                                  # A, C, G, T, then the mask

def corrupt(x0, t):
    alpha = 1.0 - t                       # linear schedule
    keep = torch.rand(x0.shape, device=DEV) < alpha
    return torch.where(keep, x0, MASK)

def loss(model, x0):
    t = torch.rand(x0.shape[0], device=DEV)
    zt = corrupt(x0, t[:, None])
    logits = model(zt, t)                 # [B, L, 5]
    logits[..., MASK] = -1e9              # no mass on the mask
    ce = F.cross_entropy(logits.transpose(1, 2), x0,
                         reduction='none')
    w = 1.0 / t                           # -alpha_dot/(1 - alpha)
    return (w[:, None] * ce * (zt == MASK)).sum(-1).mean()

@torch.no_grad()
def sample(model, B, L, steps=32):
    z = torch.full((B, L), MASK, device=DEV)
    ts = torch.linspace(1.0, 1e-3, steps + 1, device=DEV)
    for t, s in zip(ts[:-1], ts[1:]):
        logits = model(z, t.expand(B))
        logits[..., MASK] = -1e9
        x0 = Categorical(logits=logits).sample()
        unmask = torch.rand(z.shape, device=DEV) < (t - s) / t
        z = torch.where((z == MASK) & unmask, x0, z)
    return z
```

To corrupt, we sample the forward marginal directly

- The mask is token 4, the extra absorbing state.
- Each position survives with probability $\alpha_t = 1 - t$, independently.
- By fact 2, we jump to any noise level in one line, with no trajectory.

```
MASK = 4                                # A, C, G, T, then the mask

def corrupt(x0, t):
    alpha = 1.0 - t                     # linear schedule
    keep = torch.rand(x0.shape, device=DEV) < alpha
    return torch.where(keep, x0, MASK)

def loss(model, x0):
    t = torch.rand(x0.shape[0], device=DEV)
    zt = corrupt(x0, t[:, None])
    logits = model(zt, t)               # [B, L, 5]
    logits[..., MASK] = -1e9            # no mass on the mask
    ce = F.cross_entropy(logits.transpose(1, 2), x0,
                         reduction='none')
    w = 1.0 / t                         # -alpha_dot/(1 - alpha)
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    ts = torch.linspace(1.0, 1e-3, steps + 1, device=DEV)
    for t, s in zip(ts[:-1], ts[1:]):
        logits = model(z, t.expand(B))
        logits[..., MASK] = -1e9
        x0 = Categorical(logits=logits).sample()
        unmask = torch.rand(z.shape, device=DEV) < (t - s) / t
        z = torch.where((z == MASK) & unmask, x0, z)
    return z
```

To train, we sample one time and one corruption

- We sample t uniformly and corrupt the clean batch once.

```
MASK = 4                                # A, C, G, T, then the mask

def corrupt(x0, t):
    alpha = 1.0 - t                     # linear schedule
    keep = torch.rand(x0.shape, device=DEV) < alpha
    return torch.where(keep, x0, MASK)

def loss(model, x0):
    t = torch.rand(x0.shape[0], device=DEV)
    zt = corrupt(x0, t[:, None])
    logits = model(zt, t)               # [B, L, 5]
    logits[..., MASK] = -1e9            # no mass on the mask
    ce = F.cross_entropy(logits.transpose(1, 2), x0,
                         reduction='none')
    w = 1.0 / t                         # -alpha_dot/(1 - alpha)
    return (w[:, None] * ce * (zt == MASK)).sum(-1).mean()

@torch.no_grad()
def sample(model, B, L, steps=32):
    z = torch.full((B, L), MASK, device=DEV)
    ts = torch.linspace(1.0, 1e-3, steps + 1, device=DEV)
    for t, s in zip(ts[:-1], ts[1:]):
        logits = model(z, t.expand(B))
        logits[..., MASK] = -1e9
        x0 = Categorical(logits=logits).sample()
        unmask = torch.rand(z.shape, device=DEV) < (t - s) / t
        z = torch.where((z == MASK) & unmask, x0, z)
    return z
```

The denoiser predicts the clean token at every position

- We sample t uniformly and corrupt the clean batch once.
- The network reads the whole noisy sequence and returns five logits per position.

```
MASK = 4                                # A, C, G, T, then the mask

def corrupt(x0, t):
    alpha = 1.0 - t                     # linear schedule
    keep = torch.rand(x0.shape, device=DEV) < alpha
    return torch.where(keep, x0, MASK)

def loss(model, x0):
    t = torch.rand(x0.shape[0], device=DEV)
    zt = corrupt(x0, t[:, None])
    logits = model(zt, t)               # [B, L, 5]
    logits[..., MASK] = -1e9            # no mass on the mask
    ce = F.cross_entropy(logits.transpose(1, 2), x0,
                         reduction='none')
    w = 1.0 / t                         # -alpha_dot/(1 - alpha)
    return (w[:, None] * ce * (zt == MASK)).sum(-1).mean()

@torch.no_grad()
def sample(model, B, L, steps=32):
    z = torch.full((B, L), MASK, device=DEV)
    ts = torch.linspace(1.0, 1e-3, steps + 1, device=DEV)
    for t, s in zip(ts[:-1], ts[1:]):
        logits = model(z, t.expand(B))
        logits[..., MASK] = -1e9
        x0 = Categorical(logits=logits).sample()
        unmask = torch.rand(z.shape, device=DEV) < (t - s) / t
        z = torch.where((z == MASK) & unmask, x0, z)
    return z
```

The denoiser never predicts a mask

- We sample t uniformly and corrupt the clean batch once.
- The network reads the whole noisy sequence and returns five logits per position.
- This is the zero-mask rule. The posterior over clean tokens has no mass on m.

```
MASK = 4                                # A, C, G, T, then the mask

def corrupt(x0, t):
    alpha = 1.0 - t                     # linear schedule
    keep = torch.rand(x0.shape, device=DEV) < alpha
    return torch.where(keep, x0, MASK)

def loss(model, x0):
    t = torch.rand(x0.shape[0], device=DEV)
    zt = corrupt(x0, t[:, None])
    logits = model(zt, t)               # [B, L, 5]
    logits[..., MASK] = -1e9            # no mass on the mask
    ce = F.cross_entropy(logits.transpose(1, 2), x0,
                         reduction='none')
    w = 1.0 / t                         # -alpha_dot/(1 - alpha)
    return (w[:, None] * ce * (zt == MASK)).sum(-1).mean()

@torch.no_grad()
def sample(model, B, L, steps=32):
    z = torch.full((B, L), MASK, device=DEV)
    ts = torch.linspace(1.0, 1e-3, steps + 1, device=DEV)
    for t, s in zip(ts[:-1], ts[1:]):
        logits = model(z, t.expand(B))
        logits[..., MASK] = -1e9
        x0 = Categorical(logits=logits).sample()
        unmask = torch.rand(z.shape, device=DEV) < (t - s) / t
        z = torch.where((z == MASK) & unmask, x0, z)
    return z
```

We score the correct clean token with cross-entropy

- The rate error at a masked position reduced to this one term.

```
MASK = 4                                # A, C, G, T, then the mask

def corrupt(x0, t):
    alpha = 1.0 - t                     # linear schedule
    keep = torch.rand(x0.shape, device=DEV) < alpha
    return torch.where(keep, x0, MASK)

def loss(model, x0):
    t = torch.rand(x0.shape[0], device=DEV)
    zt = corrupt(x0, t[:, None])
    logits = model(zt, t)               # [B, L, 5]
    logits[..., MASK] = -1e9            # no mass on the mask
    ce = F.cross_entropy(logits.transpose(1, 2), x0,
                         reduction='none')
    w = 1.0 / t                         # -alpha_dot/(1 - alpha)
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@torch.no_grad()
def sample(model, B, L, steps=32):
    z = torch.full((B, L), MASK, device=DEV)
    ts = torch.linspace(1.0, 1e-3, steps + 1, device=DEV)
    for t, s in zip(ts[:-1], ts[1:]):
        logits = model(z, t.expand(B))
        logits[..., MASK] = -1e9
        x0 = Categorical(logits=logits).sample()
        unmask = torch.rand(z.shape, device=DEV) < (t - s) / t
        z = torch.where((z == MASK) & unmask, x0, z)
    return z
```

The weight is the reveal rate w(t)

- The rate error at a masked position reduced to this one term.
- With $\alpha_t = 1 - t$, the weight $-\dot{\alpha}_t/(1 - \alpha_t)$ is exactly $1/t$.

```
MASK = 4                              # A, C, G, T, then the mask

def corrupt(x0, t):
    alpha = 1.0 - t                   # linear schedule
    keep = torch.rand(x0.shape, device=DEV) < alpha
    return torch.where(keep, x0, MASK)

def loss(model, x0):
    t = torch.rand(x0.shape[0], device=DEV)
    zt = corrupt(x0, t[:, None])
    logits = model(zt, t)             # [B, L, 5]
    logits[..., MASK] = -1e9          # no mass on the mask
    ce = F.cross_entropy(logits.transpose(1, 2), x0,
                         reduction='none')
    w = 1.0 / t                       # -alpha_dot/(1 - alpha)
    return (w[:, None] * ce * (zt == MASK)).sum(-1).mean()

@torch.no_grad()
def sample(model, B, L, steps=32):
    z = torch.full((B, L), MASK, device=DEV)
    ts = torch.linspace(1.0, 1e-3, steps + 1, device=DEV)
    for t, s in zip(ts[:-1], ts[1:]):
        logits = model(z, t.expand(B))
        logits[..., MASK] = -1e9
        x0 = Categorical(logits=logits).sample()
        unmask = torch.rand(z.shape, device=DEV) < (t - s) / t
        z = torch.where((z == MASK) & unmask, x0, z)
    return z
```

Only masked positions contribute

- The rate error at a masked position reduced to this one term.
- With $\alpha_t = 1 - t$, the weight $-\dot{\alpha}_t/(1 - \alpha_t)$ is exactly $1/t$.
- Visible positions have zero rate in both processes, so their loss is zero.

```python
MASK = 4                                # A, C, G, T, then the mask

def corrupt(x0, t):
    alpha = 1.0 - t                     # linear schedule
    keep = torch.rand(x0.shape, device=DEV) < alpha
    return torch.where(keep, x0, MASK)

def loss(model, x0):
    t = torch.rand(x0.shape[0], device=DEV)
    zt = corrupt(x0, t[:, None])
    logits = model(zt, t)               # [B, L, 5]
    logits[..., MASK] = -1e9            # no mass on the mask
    ce = F.cross_entropy(logits.transpose(1, 2), x0,
                         reduction='none')
    w = 1.0 / t                         # -alpha_dot/(1 - alpha)
    return (w[:, None] * ce * (zt == MASK)).sum(-1).mean()

@torch.no_grad()
def sample(model, B, L, steps=32):
    z = torch.full((B, L), MASK, device=DEV)
    ts = torch.linspace(1.0, 1e-3, steps + 1, device=DEV)
    for t, s in zip(ts[:-1], ts[1:]):
        logits = model(z, t.expand(B))
        logits[..., MASK] = -1e9
        x0 = Categorical(logits=logits).sample()
        unmask = torch.rand(z.shape, device=DEV) < (t - s) / t
        z = torch.where((z == MASK) & unmask, x0, z)
    return z
```

Generation starts where the forward process ends

- All masks is the stationary distribution of the masking rates.

```
MASK = 4                                # A, C, G, T, then the mask

def corrupt(x0, t):
    alpha = 1.0 - t                     # linear schedule
    keep = torch.rand(x0.shape, device=DEV) < alpha
    return torch.where(keep, x0, MASK)

def loss(model, x0):
    t = torch.rand(x0.shape[0], device=DEV)
    zt = corrupt(x0, t[:, None])
    logits = model(zt, t)               # [B, L, 5]
    logits[..., MASK] = -1e9            # no mass on the mask
    ce = F.cross_entropy(logits.transpose(1, 2), x0,
                         reduction='none')
    w = 1.0 / t                         # -alpha_dot/(1 - alpha)
    return (w[:, None] * ce * (zt == MASK)).sum(-1).mean()

@torch.no_grad()
def sample(model, B, L, steps=32):
    z = torch.full((B, L), MASK, device=DEV)
    ts = torch.linspace(1.0, 1e-3, steps + 1, device=DEV)
    for t, s in zip(ts[:-1], ts[1:]):
        logits = model(z, t.expand(B))
        logits[..., MASK] = -1e9
        x0 = Categorical(logits=logits).sample()
        unmask = torch.rand(z.shape, device=DEV) < (t - s) / t
        z = torch.where((z == MASK) & unmask, x0, z)
    return z
```

At each step, we draw a clean token from the denoiser

- All masks is the stationary distribution of the masking rates.
- This sample is the posterior average turned into one reverse step.

```python
MASK = 4                                # A, C, G, T, then the mask

def corrupt(x0, t):
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```

Each masked position reveals with probability (t – s)/t

- All masks is the stationary distribution of the masking rates.
- This sample is the posterior average turned into one reverse step.
- That is the reveal rate w(t) integrated over the step, α_s/α_t apart from one.

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```

Revealed tokens never change again

- This is carry-over unmasking, which we derived from the rates being zero.

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```

Thirty lines hold the entire framework for masking

- This is carry-over unmasking, which we derived from the rates being zero.

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```

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For block diffusion, the loop runs over blocks, and the denoiser also reads the clean prefix.

In the rest of this lecture, we change how we choose the next position, how we weight the loss, and what we condition on.

Let’s start with the choice of position

For uniform noise, we corrupt with a replacement token instead of the mask, and we use all K – 1 rates in the reverse step.

For block diffusion, the loop runs over blocks, and the denoiser also reads the clean prefix.

In the rest of this lecture, we change how we choose the next position, how we weight the loss, and what we condition on.

If the model is more confident at some positions, why reveal them in a random order?

P2: Path Planning

Path Planning for Diffusion Language Model Sampling

Fred Zhangzhi Peng[1,*,‡], **Zachary Bezemek**[1,*], **Sawan Patel**[2], **Jarrid Rector-Brooks**[3,4], **Sherwood Yao**[2], **Avishek Joey Bose**[3,5], **Alexander Tong**[3,4,6,†,‡], **Pranam Chatterjee**[7,†,‡]

[1]Duke University, [2]Atom Bioworks, [3]Mila – Québec AI Institute,
[4]Université de Montréal, [5]The University of Oxford, [6]Aithyra, [7]University of Pennsylvania
[*]Equal contribution, [†]Equal co-supervision
[‡]Corresponding authors: zp70@duke.edu, atong@aithyra.at, pranam@seas.upenn.edu

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The corrector rate matrix C_t equals the forward rates R_t plus the reverse rates $\hat{R}_t$ at the same time t.

Masking a token again and revealing it is one such step.

Here is one sampling step of P2 in code

```
def p2_step(model, z, t, s, n_unmask, n_remask):
    logits = model(z, t)
    logits[..., MASK] = -1e9
    p = logits.softmax(-1)
    x0 = Categorical(probs=p).sample()

    # planner score: how sure the denoiser is here
    score = p.max(-1).values                # or -entropy, or a
                                            # separate planner
    masked = (z == MASK)
    pick = topk_where(score, masked, n_unmask)
    z = torch.where(pick, x0, z)            # unmask the best

    # corrector: send the least certain visible tokens back
    back = topk_where(-score, ~masked, n_remask)
    z = torch.where(back, MASK, z)          # remask the worst
    return z
```

First, look at where the denoiser is confident

- From one forward pass, we get a distribution at every masked position.

```
def p2_step(model, z, t, s, n_unmask, n_remask):
    logits = model(z, t)
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    return z
```

Next, rank the masked positions by the planner score

- From one forward pass, we get a distribution at every masked position.
- In P2, we score positions by confidence, by entropy, or with a separate planner.

```
def p2_step(model, z, t, s, n_unmask, n_remask):
    logits = model(z, t)
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```

We reveal the positions we are most sure about

- The reverse rates are unchanged, and only the order of reveals changes.

```
def p2_step(model, z, t, s, n_unmask, n_remask):
    logits = model(z, t)
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    back = topk_where(-score, ~masked, n_remask)
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    return z
```

In a corrector step, uncertain tokens go back to the mask

- The reverse rates are unchanged, and only the order of reveals changes.
- Masking a token again and revealing it is a corrector step, and p_t is stationary for it.

```python
def p2_step(model, z, t, s, n_unmask, n_remask):
    logits = model(z, t)
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```

So we can plan and remask without retraining the model

- The reverse rates are unchanged, and only the order of reveals changes.
- Masking a token again and revealing it is a corrector step, and p_t is stationary for it.

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def p2_step(model, z, t, s, n_unmask, n_remask):
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P2: a reveal and a correction

Toy state A [MASK] G [MASK]. The planner scores positions 1 through 4 as (0.55, 0.90, 0.80, 0.60). Reveal one masked position, then remask one previously visible position.

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Among the two previously visible positions, position 1 has the lower score, so it becomes a mask.

The planner chooses where to spend the next denoising step, including where to reconsider a decision.

We select the positions with a planner, and we remask to correct earlier choices

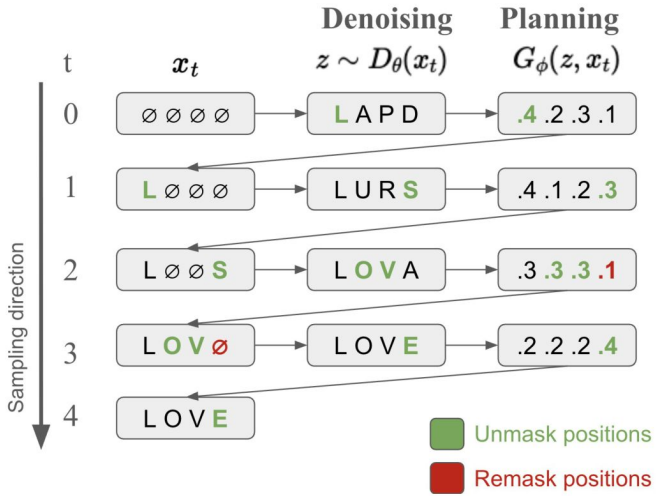


Figure 1: Illustration of P2 sampling (Algorithm 1). At each step, the denoiser D_θ predicts z, and the planner G_ϕ selects positions to unmask (green) and remask (red).

We can sample any MDLM checkpoint this way.

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Here is the protein benchmark in Table 2 of the paper

Table 2: Protein sequence generation benchmark. We evaluate structure quality via pLDDT, pTM, and pAE, and diversity via token entropy and sequence uniqueness. Foldability is the percentage of sequences satisfying pLDDT > 80, pTM > 0.7, and pAE < 10. See Section F.1 for setup and Table S4 for model size ablations.

Model	pLDDT↑	pTM↑	pAE↓	Foldability (%)↑	Entropy↑	Diversity (%)↑
EvoDiff	31.84	0.21	24.76	0.43	**4.05**	93.19
ESM3	34.13	0.23	24.65	1.50	3.99	**93.44**
ProGen2	49.38	0.28	23.38	4.48	2.55	89.31
DPLM	80.23	0.65	12.07	48.14	3.14	92.80
DPLM + P2-train (ours)	**83.45**	**0.72**	**10.15**	**58.86**	3.35	92.69

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A masked diffusion protein model folds 48.1% of its samples

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Peng,..,Chatterjee. *TMLR* (2026)

When we plan the path, that rises to 58.9%, with pLDDT from 80.2 to 83.5

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Peng,..,Chatterjee. *TMLR* (2026)

Only the order of reveals changed, with the same weights

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Peng,..,Chatterjee. *TMLR* (2026)

If we change the order of generation, shouldn't we change training too?

PAPL: Planner-Aware Path Learning

PLANNER AWARE PATH LEARNING IN DIFFUSION LANGUAGE MODELS TRAINING

Fred Zhangzhi Peng[1,‡,*], **Zachary Bezemek**[1,‡,*], **Jarrid Rector-Brooks**[2,3,4], **Shuibai Zhang**[5], **Anru R. Zhang**[1], **Michael Bronstein**[6,7], **Alexander Tong**[7†], **Avishek Joey Bose**[2,6,8†]

[1] Duke University [2] Mila [3] Université de Montréal [4] California Institute of Technology
[5] University of Wisconsin–Madison [6] University of Oxford [7] AITHYRA
[8] Imperial College London [‡] Equal contribution [†] Equal advising

Peng,..,Bose. *ICLR* (2026)

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In PAPL, we put the planner's weights inside the rate loss:

$$\mathcal{L}_{\mathrm{PAPL}} = \int_0^1 w(t)\, \mathbb{E}\Big[\sum_{i:\, Z_t^i = m} \lambda_i(Z_t) \left(-\log d_\theta^i(x_0^i \mid Z_t, t) \right) \Big] dt$$

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$$\mathcal{L}_{\mathrm{PAPL}} = \int_0^1 w(t)\, \mathbb{E}\Big[\sum_{i:\, Z_t^i = m} \lambda_i(Z_t) \left(-\log d_\theta^i(x_0^i \mid Z_t, t) \right) \Big] dt$$

The PAPL loss L_{PAPL} equals the integral over time of the weight w(t) times the expected sum over masked positions i of the planner weight $\lambda_i(Z_t)$ times negative log $d^i_\theta(x_0{}^i \mid Z_t, t)$, the negative log probability of the clean token $x_0{}^i$.

Next, we bring the planner into training

MDLM's loss weights every masked position equally, because we reveal them in a uniform random order.

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With uniform weights $\lambda_i = 1$, this is the MDLM loss we derived.

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If a planner reveals a position often, mistakes at that position receive more weight during training.

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The five facts still hold, with the same rates, the same marginals, and the same bound with a different weighting.

PAPL: the planner changes the training weight

Toy state A [MASK] G [MASK], with target C and T at the masks. The denoiser gives them probabilities 0.5 and 0.25. Compare equal weights with planner weights 0.8 and 0.2 at a fixed time.

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Training emphasizes the positions that this planner is likely to reveal next.

We reweight the rate loss by the planner's choice of path

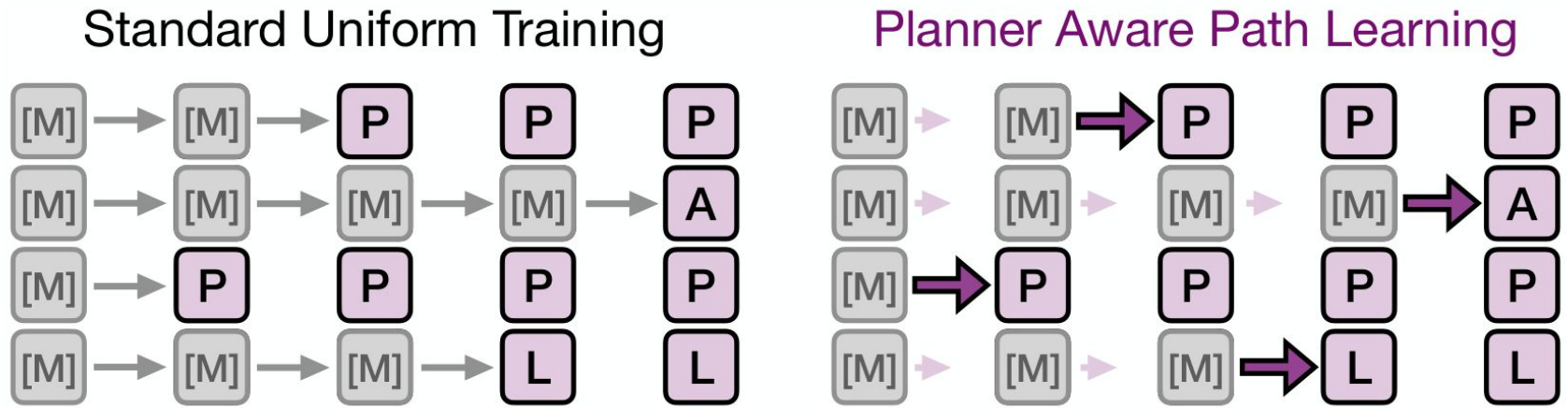


Figure 1: **Planner-Aware Path Learning (PAPL) resolves training–inference mismatch in DLMs.** Standard uniform training for DLMs (left) applies a uniform loss across all masked positions, distributing capacity over regions that inference-time planners never traverse. PAPL (right) introduces planner-aware weights into the loss, aligning training with the planner's preferred trajectories (outlined arrows) and eliminating training-inference mismatch.

Peng,..,Bose. *ICLR* (2026)

Here is the protein benchmark again, in Table 1 of the paper

Table 1: Protein sequence generation benchmark. We evaluate structure quality via pLDDT, pTM, and pAE, and diversity via token entropy and sequence uniqueness. Foldability is the percentage of sequences satisfying pLDDT > 80, pTM > 0.7, and pAE < 10.

	Model	pLDDT↑	pTM↑	pAE↓	Foldability (%)↑	Entropy↑	Diversity (%)↑
Large	ESM3	34.13	0.23	24.65	1.50	3.99	**93.44**
	ProGen2-medium	57.94	0.38	20.81	12.75	2.91	91.45
	ProGen2-large	55.07	0.35	22.00	11.87	2.73	91.48
	DPLM-650M	79.53	0.66	11.85	49.14	3.18	92.22
150M-scale	EvoDiff	31.84	0.21	24.76	0.43	**4.05**	93.19
	ProGen2-small	49.38	0.28	23.38	4.48	2.55	89.31
	DPLM-150M	80.23	0.65	12.07	48.14	3.14	92.80
	DLM-150M	81.32	0.65	12.00	42.43	3.21	92.45
	DLM-150M + PAPL (ours)	**81.48**	**0.72**	**8.97**	**59.40**	3.12	91.73

Peng,..,Bose. *ICLR* (2026)

The same 150M diffusion model folds 42.4% of its samples

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Peng,..,Bose. *ICLR* (2026)

With planner-aware training, that rises to 59.4%, and pAE falls from 12.00 to 8.97

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Peng,..,Bose. *ICLR* (2026)

PAPL is a reweighted rate loss, with the same rates and the same bound

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Peng,..,Bose. *ICLR* (2026)

How can we steer these samples toward peptides that bind, dissolve, and stay stable?

PepTune

PepTune: *De Novo* Generation of Therapeutic Peptides with Multi-Objective-Guided Discrete Diffusion

Sophia Tang [*1] **Yinuo Zhang** [*2] **Pranam Chatterjee** [1 3]

Tang,..,Chatterjee. *ICML* (2025)

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So far, we have modeled the data distribution. For therapeutic design, we also need peptides to bind, stay soluble, and cross a membrane.

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Recall from Doob's h-transform how to condition a jump process on a property, with $h_t(x) = p(c \mid X_t = x)$.

In PepTune, we estimate that value by rolling out the denoiser to complete sequences and scoring them.

PepMDLM is masked diffusion over peptide SMILES, with every token a chemical fragment

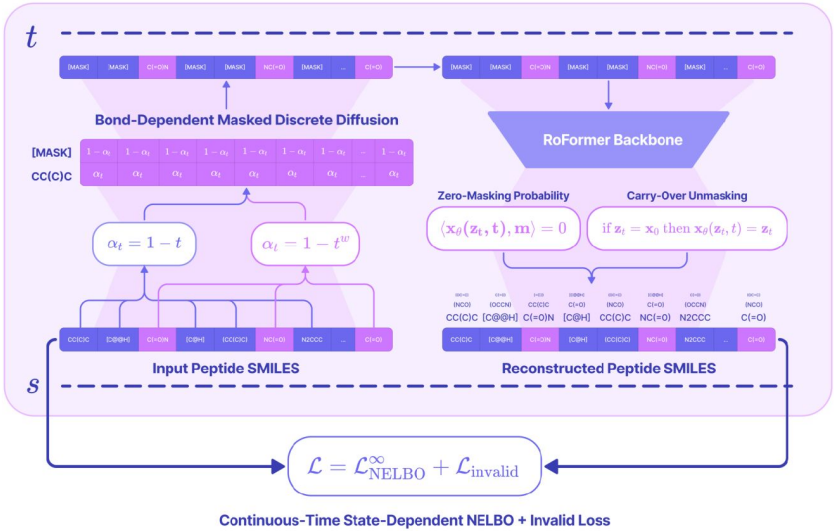


Figure 1. **PepMDLM**. PepMDLM is a discrete masked diffusion model for unconditional *de novo* generation of peptide SMILES representations.

Tang,..,Chatterjee. *ICML* (2025)

Next, we estimate the value h with Monte Carlo tree search

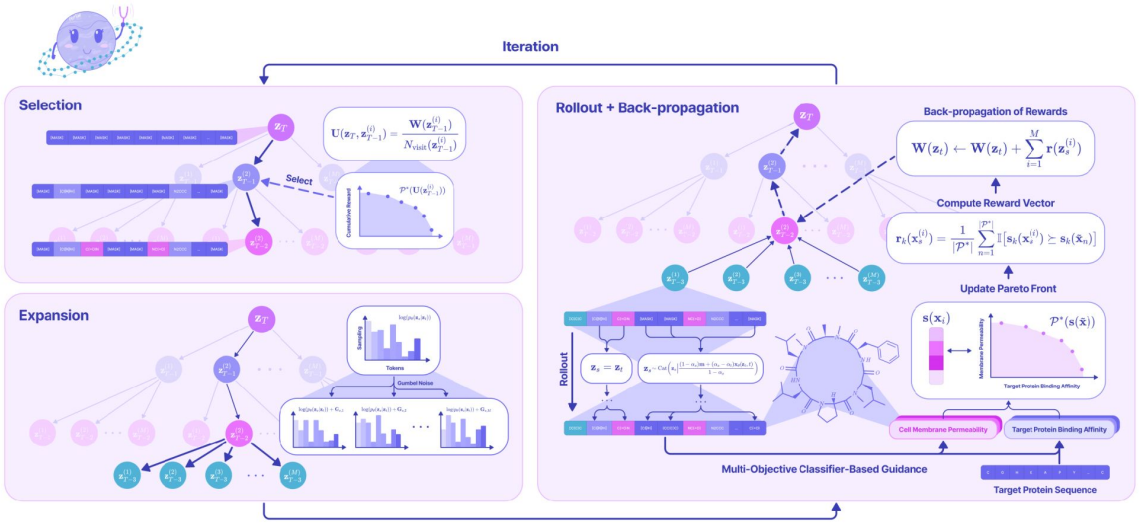


Figure 2. **PepTune**. PepTune is a multi-objective discrete diffusion model with Monte Carlo Tree Guidance (MCTG). The full algorithm is detailed in Algorithm 3.

Tang,..,Chatterjee. *ICML* (2025)

In selection, we follow a branch, balancing reward against how often it was explored

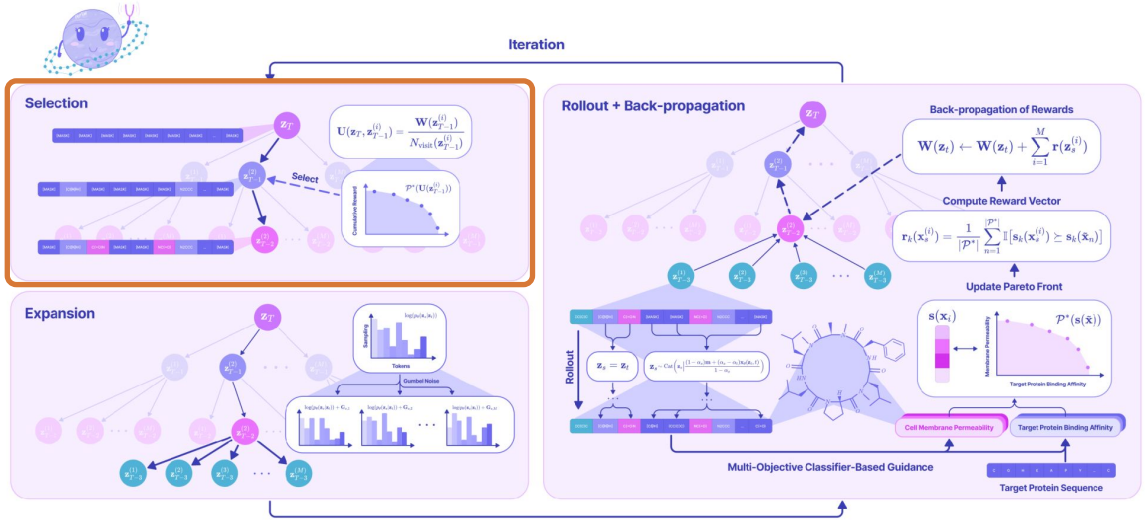


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Tang,..,Chatterjee. *ICML* (2025)

In expansion, the next unmasking comes from the denoiser, with Gumbel noise

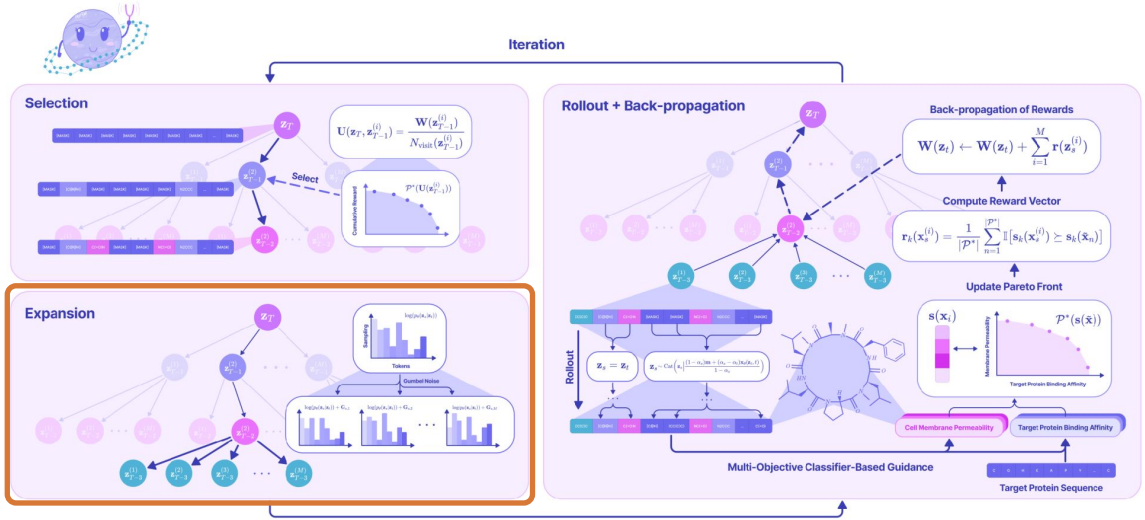


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In each rollout, we finish the peptide and score it on every objective

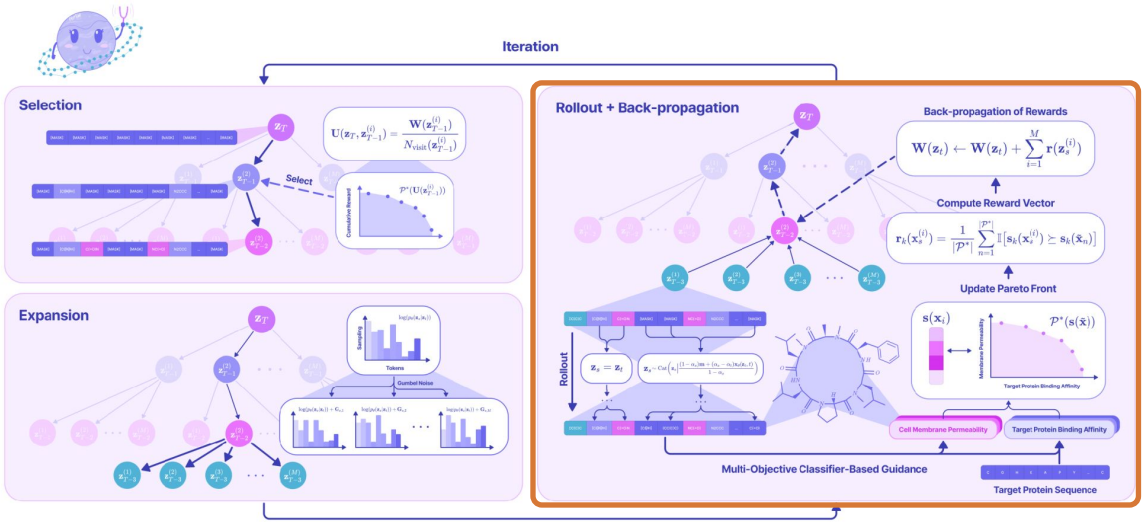


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Those scores estimate h_t, and the rewards flow back up the tree

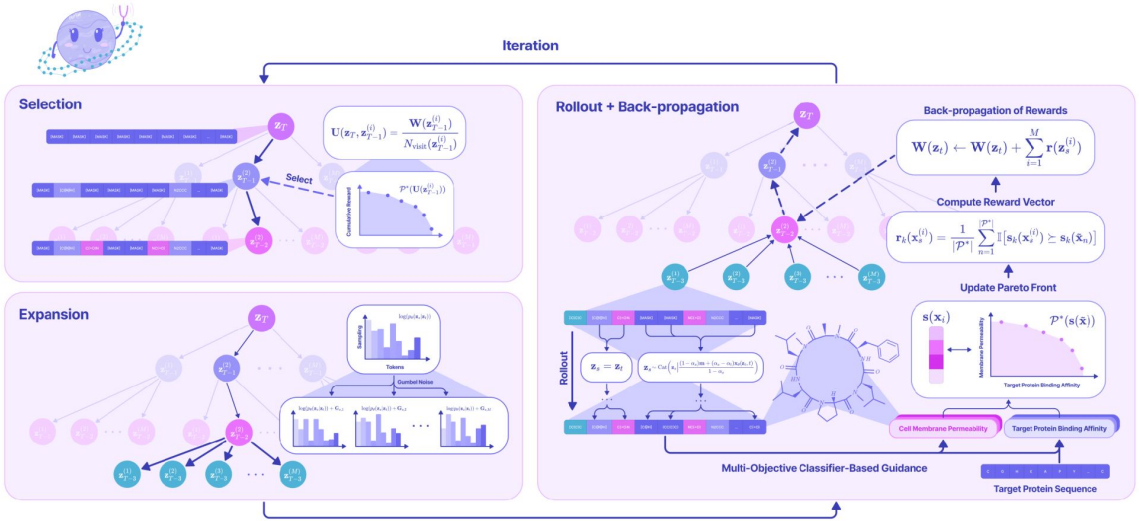


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Now write the rollout estimate of the value

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In each rollout, we complete z and score it, so the average reward estimates the value:

$$h_t(z) \approx \frac{1}{N} \sum_{n=1}^{N} r\left(x_0^{(n)}\right), \qquad x_0^{(n)} \sim p_\theta(\cdot \mid z)$$

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The value $h_t(z)$ is approximately the average over N rollouts of the reward $r(x_0^{(n)})$, where each completed sequence $x_0^{(n)}$ is sampled from the model $p_\theta(\cdot \mid z)$ starting from z.

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Then apply the h-transform to each reverse jump: $\hat{R}_t^c(z, z') = \hat{R}_t(z, z') \, \dfrac{h_t(z')}{h_t(z)}$

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The guided reverse rate $\hat{R}^c_t(z, z')$ from z to z′ equals the model's reverse rate $\hat{R}_t(z, z')$ times the estimated value $h_t(z')$ of z′ divided by the value $h_t(z)$ of z.

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In selection, we balance reward against how often a branch has been explored, which is the usual tree-search trade-off.

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A partial peptide gets a high value when its sampled completions tend to have the properties we want.

PepTune: two rollouts change a jump rate

Use a toy scalar reward for two candidate peptide edits P and Q. Their rollout rewards are (0.2, 0.8) and (0.4, 0.4). Base rates are 2 and 1, and the current estimated value is 0.45.

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$$R^c(P) = 2\frac{0.5}{0.45} \approx 2.222, \qquad R^c(Q) = 1\frac{0.4}{0.45} \approx 0.889$$

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A partially generated peptide gets more sampling effort when its continuations look promising.

Here is the peptide quality in Table 1 of the paper

Table 1. **Evaluation metrics for generative quality of peptide SMILES sequences of max token length set to 200.**

Model	Validity (↑)	Uniqueness (↑)	Diversity (↑)	SNN (↓)	Randomness (↑)	KL-Divergence (↑)
Data	1.000	1.000	0.885	1.000	4.55	0 (Reference)
PepMDLM	0.450	1.000	0.705	0.513	4.11	0.174
PepTune	1.000	1.000	0.677	0.486	4.12	0.173

Tang,..,Chatterjee. *ICML* (2025)

The unguided peptide model produces valid SMILES 45% of the time

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Searching under the h-transform reaches 100% validity at the same diversity

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Model	Validity (↑)	Uniqueness (↑)	Diversity (↑)	SNN (↓)	Randomness (↑)	KL-Divergence (↑)
Data	1.000	1.000	0.885	1.000	4.55	0 (Reference)
PepMDLM	0.450	1.000	0.705	0.513	4.11	0.174
PepTune	1.000	1.000	0.677	0.486	4.12	0.173

Tang,..,Chatterjee. *ICML* (2025)

The model is unchanged, and only its reverse rates are reweighted

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If search finds better generation paths, could the model learn to follow them on its own?

TR2-D2

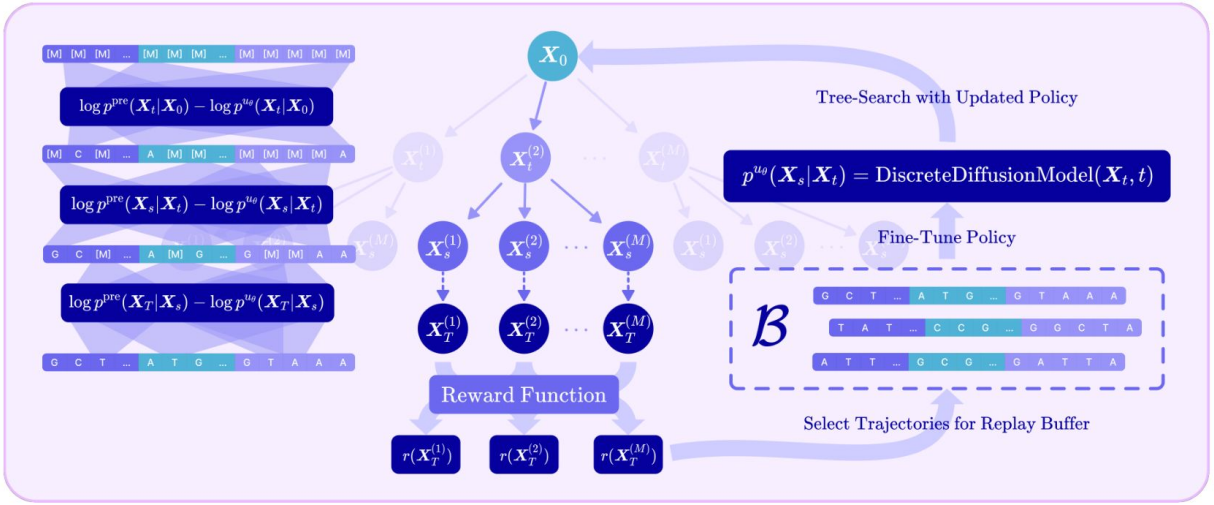


Figure 1: **Tree Search Guided Trajectory-Aware Fine-Tuning for Discrete Diffusion.** Our framework has two key components: **(1) a tree search algorithm** to generate a replay buffer of diffusion trajectories optimized for one or more reward functions using the current policy and **(2) an off-policy RL algorithm** for discrete diffusion fine-tuning using the optimized replay buffer.

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This is the h-transform again, with $h(x) = \exp(r(x)/\lambda)$ at the end of generation.

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Lower temperature makes a fixed reward difference change the target probabilities more strongly.

We explore reverse paths of the current model by tree search

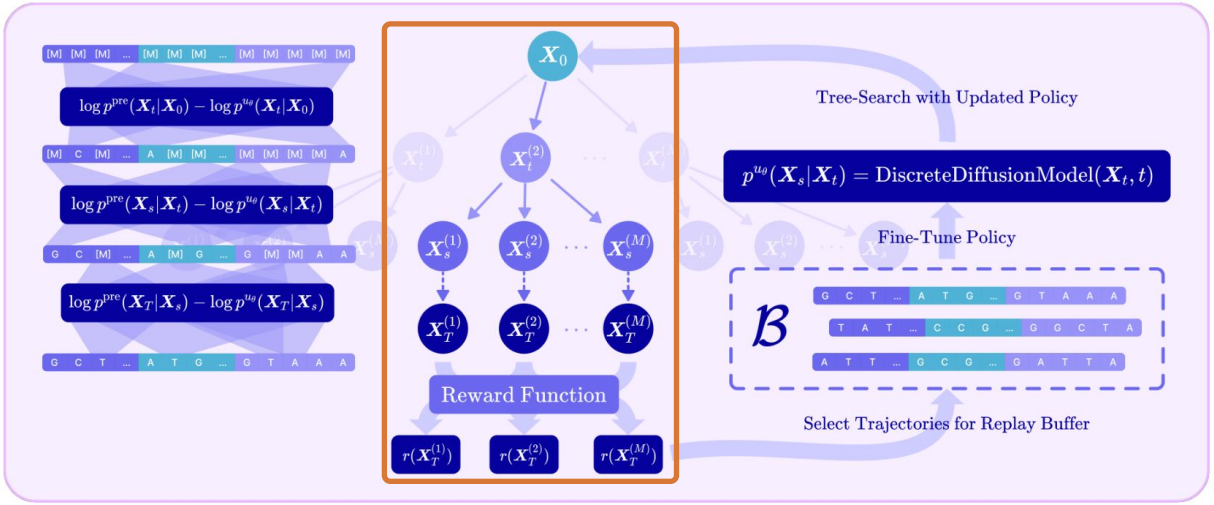


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Each completed path has a weight, the reward against its path probability

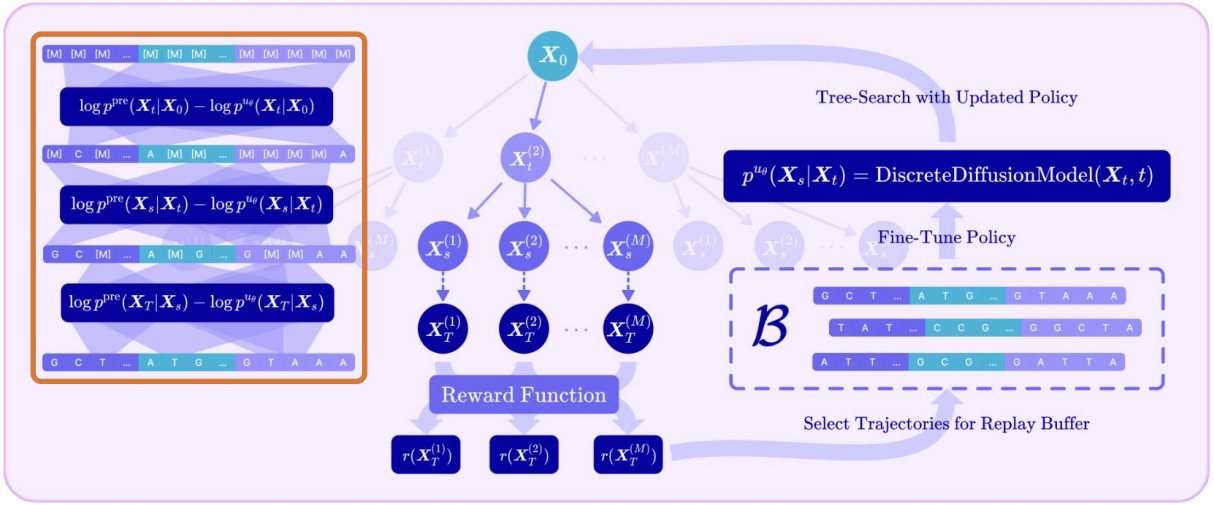


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When we train on those weighted paths, the model moves toward the tilted target

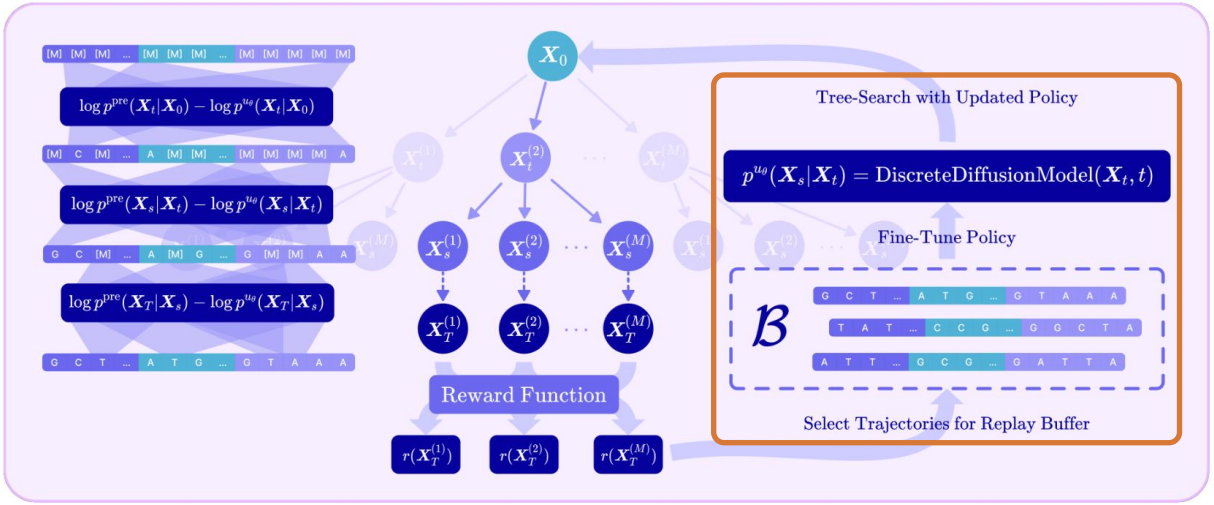


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The weights are products of the rate ratios we have been computing

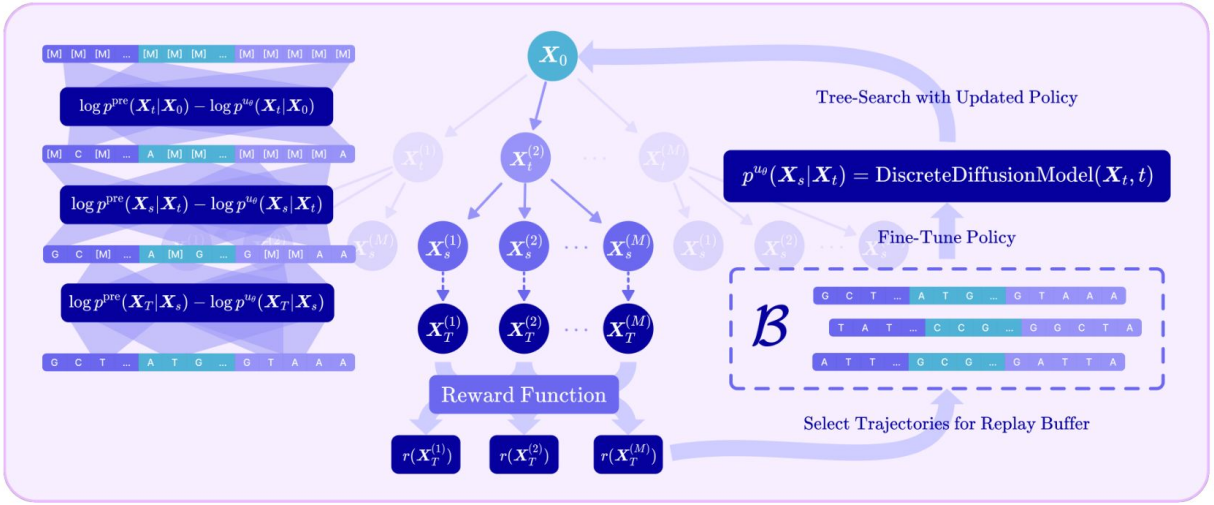


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Training on weighted search trajectories aims to make this preference available in the model itself.

Here is the regulatory DNA result in Table 1 of the paper

Table 1: **Comparison of TR2-D2 for regulatory DNA generation optimized on enhancer activity.** Metrics were computed for 640 sequences across 3 seeds, with standard deviations reported. Best values are **bolded**. Evaluation metrics are detailed in Appendix D.

Method	Pred-Activity (median; ↑)	ATAC-Acc (%; ↑)	3-mer Corr (↑)	App-Log-Lik (median; ↑)
pre-trained	$0.17_{\pm 0.04}$	$1.5_{\pm 0.2}$	$-0.061_{\pm 0.034}$	$-261_{\pm 0.6}$
CG	$3.30_{\pm 0.00}$	$0.0_{\pm 0.0}$	$-0.065_{\pm 0.001}$	$-266_{\pm 0.6}$
SMC	$4.15_{\pm 0.33}$	$39.9_{\pm 8.7}$	$0.840_{\pm 0.045}$	$-259_{\pm 2.5}$
TDS	$4.64_{\pm 0.21}$	$45.3_{\pm 16.4}$	$0.848_{\pm 0.008}$	$-257_{\pm 1.5}$
CFG	$5.04_{\pm 0.06}$	$92.1_{\pm 0.9}$	$0.746_{\pm 0.001}$	$-265_{\pm 0.6}$
DRAKES	$5.61_{\pm 0.07}$	$92.5_{\pm 0.6}$	$0.887_{\pm 0.002}$	$-264_{\pm 0.6}$
SEPO	$7.55_{\pm 0.01}$	$99.5_{\pm 0.2}$	$0.500_{\pm 0.004}$	$-243.8_{\pm 0.5}$
GLID^{2}E	$7.35_{\pm 0.07}$	$90.6_{\pm 0.3}$	$0.490_{\pm 0.074}$	$\mathbf{-239.9_{\pm 1.4}}$
TR2-D2 ($\alpha = 0.1$)	$6.56_{\pm 0.02}$	$86.9_{\pm 1.18}$	$\mathbf{0.925_{\pm 0.002}}$	$-259.4_{\pm 0.2}$
TR2-D2 ($\alpha = 0.001$)	$\mathbf{9.74_{\pm 0.01}}$	$\mathbf{99.9_{\pm 0.01}}$	$0.548_{\pm 0.001}$	$-271.8_{\pm 0.1}$

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The fine-tuned model reaches 9.74, with 99.9% chromatin accessibility

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Could the model decide the sequence length while it generates the tokens?

A2D2

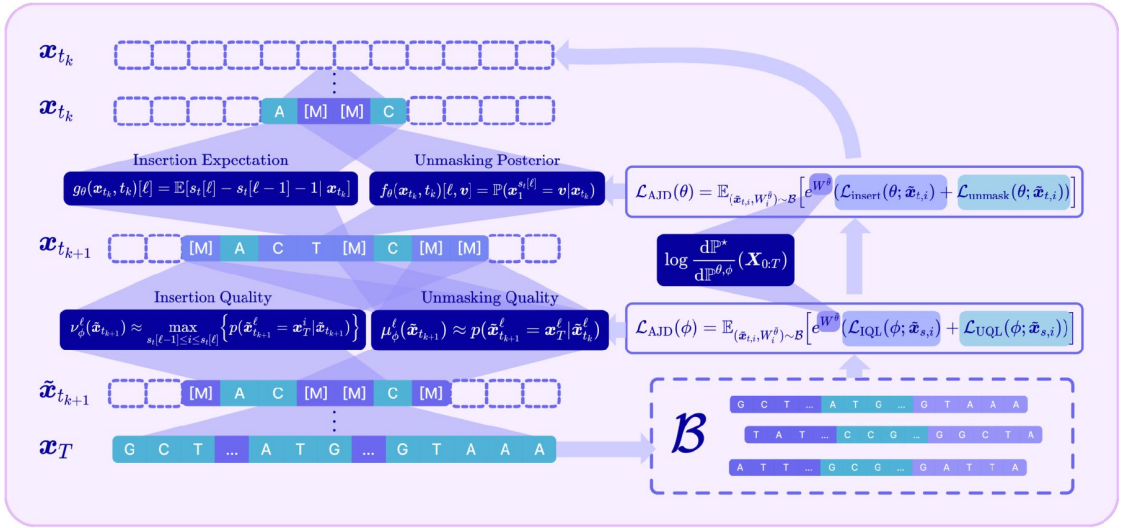


Figure 1 ***Fine-Tuning Any-Length Discrete Diffusion for Adaptive Decoding.*** *A2D2 jointly fine-tunes the insertion and unmasking policies together with a quality classifier to sample from the optimal reward-tilted distribution $\mathbb{P}^\star$. Sequences are generated via adaptive inference and stored in a replay buffer, then used to compute the Adaptive Joint Decoding loss $\mathcal{L}_{AJD}$, which weights each sequence's contribution by an importance weight $W^{\theta,\bar{\phi}}$.*

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Both are transitions in one rate matrix, so the master equation, the reversal, and the rate loss carry over.

With an insertion policy, we choose how many positions to add at each gap

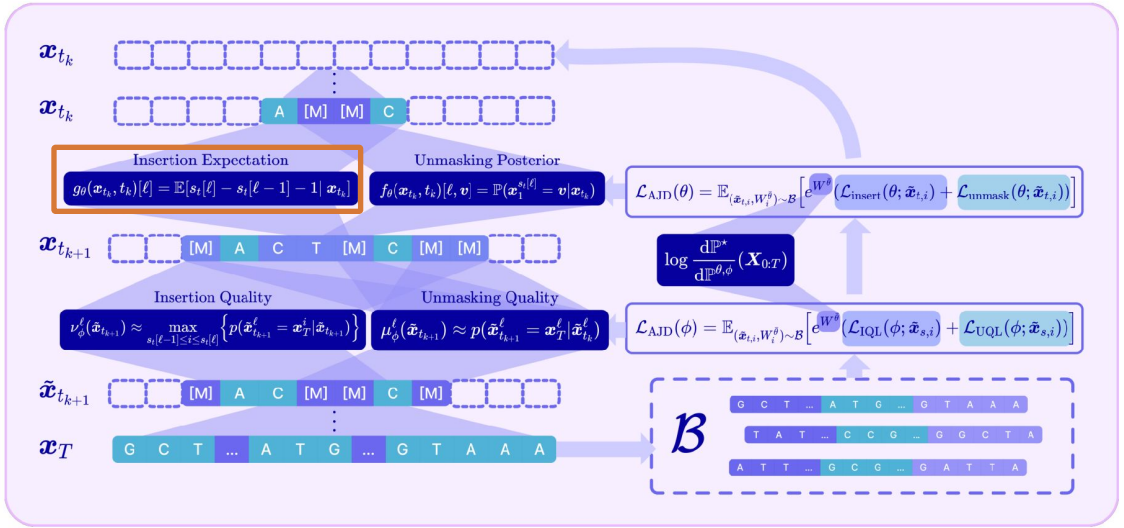


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Then fill them with an unmasking policy, as in MDLM

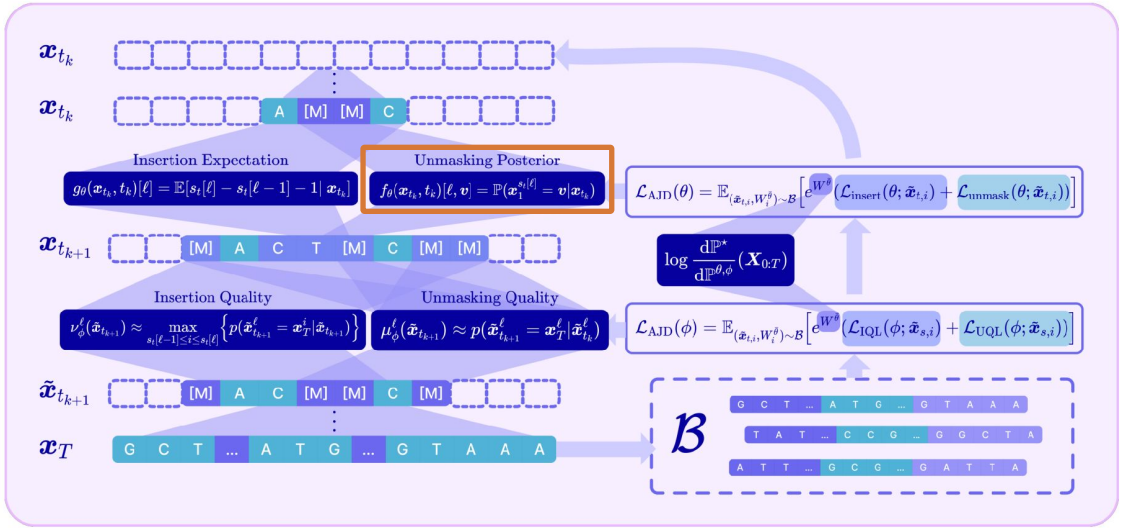


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With a quality head, we score whether to trust each decision

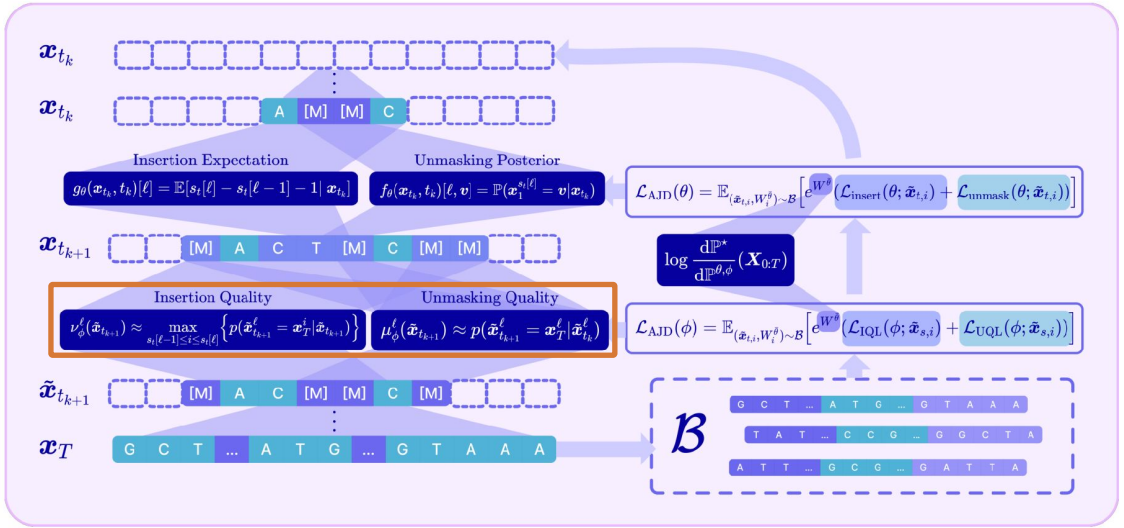


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The sequence grows and fills at the same time

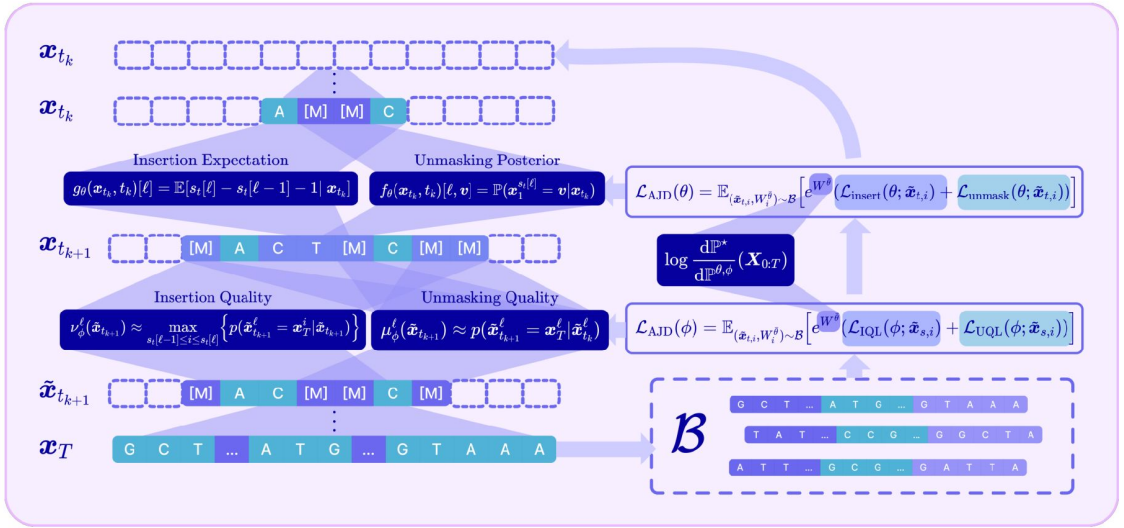


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Now write the path weight when the length can change

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With insertions, the total rate of leaving a state depends on its length. We must account for the waiting time in the path likelihood.

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The log weight log W equals the sum over the jumps along the path of the log of the model rate $\hat{R}^{\theta}$ divided by the reference rate $\hat{R}$, minus the integral over time of $\hat{R}^{\theta}_{\text{out}}$, the model's total rate of leaving the current state, minus $\hat{R}_{\text{out}}$, the reference total rate.

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That second term is the $\ell(A, B) = A \log(A/B) - A + B$ staying contribution, which matters here.

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A path includes how long nothing happens, so models with the same jump choices can still give it different probabilities.

A2D2: a path includes the waiting time

Toy path of duration 0.5 with one insertion, AG to ACG. At that jump the model rate is 2 and the reference rate is 1. Throughout this path, total exit rates are 3 and 2.

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When the sequence can grow, how long it waits carries information alongside which edit occurs.

Here is the small molecule result in Table 1 of the paper

Table 1 ***Drug-like small molecule design results.*** *Metrics were computed for* 1000 *generated sequences, and the mean and standard deviations were computed across* 3 *seeds.*

Method	Validity (% total)	Uniqueness (% valid)	Quality (% total)	QED (↑)	Synthetic Accessibility (↓)	Diversity (↑)
GenMol w/o confidence	$98.467_{\pm 0.094}$	$99.763_{\pm 0.048}$	$73.667_{\pm 1.159}$	$0.763_{\pm 0.002}$	$3.024_{\pm 0.017}$	$0.859_{\pm 0.000}$
GenMol w/ confidence	$99.733_{\pm 0.047}$	$99.532_{\pm 0.125}$	$84.000_{\pm 0.566}$	$0.813_{\pm 0.004}$	$2.881_{\pm 0.020}$	$0.817_{\pm 0.000}$
Pre-trained (Any Length)	$95.800_{\pm 0.294}$	$99.582_{\pm 0.086}$	$44.167_{\pm 0.574}$	$0.641_{\pm 0.002}$	$3.401_{\pm 0.032}$	$0.908_{\pm 0.000}$
A2D2 w/o quality	$96.400_{\pm 0.535}$	$62.484_{\pm 0.452}$	$76.067_{\pm 0.896}$	$0.795_{\pm 0.005}$	$2.306_{\pm 0.024}$	$0.730_{\pm 0.003}$
A2D2 w/o insertion quality	$92.200_{\pm 0.082}$	$93.746_{\pm 0.335}$	$71.067_{\pm 0.411}$	$0.752_{\pm 0.003}$	$2.654_{\pm 0.014}$	$0.832_{\pm 0.001}$
A2D2 w/o unmasking quality	$98.600_{\pm 0.082}$	$85.801_{\pm 0.387}$	$82.367_{\pm 0.736}$	$0.818_{\pm 0.003}$	$2.753_{\pm 0.026}$	$0.789_{\pm 0.002}$
A2D2 w/ both quality	$94.533_{\pm 0.665}$	$93.695_{\pm 0.925}$	$71.300_{\pm 0.852}$	$0.762_{\pm 0.004}$	$2.870_{\pm 0.018}$	$0.843_{\pm 0.001}$

Tang,..,Chatterjee. *arXiv* (2026)

The pre-trained any-length model reaches 44.2% quality

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After fine-tuning, that rises to 76.1%, with QED from 0.641 to 0.795

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With the quality heads, we trade some of that back for validity and diversity

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The length and the tokens come from the same rates

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	MDLM	D3PM	Uniform	Block	Simplex

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Closed-form marginal	$\alpha_t \delta_{x0} + (1 - \alpha_t)\delta_m$	$x_0 Q_t$	$\alpha_t \delta_{x0} + (1 - \alpha_t)/K$	MDLM marginal per block	mean $\alpha_t e_{x0} + (1 - \alpha_t)\pi$

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Where it ends	all masks	π	uniform tokens	all-mask block	beliefs around π
Reverse via posterior	reveal at rate $w(t)d_\theta$	Bayes step averaged over $\tilde{p}_\theta$	ratios averaged over d_θ	MDLM rates given the prefix	kernel averaged over d_θ

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Loss	weighted cross-entropy	$L_{vb} + \lambda$ CE	full rate loss	sum of block losses	cross-entropy

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If you can prove the five facts for your rate matrix, you have a discrete diffusion model.

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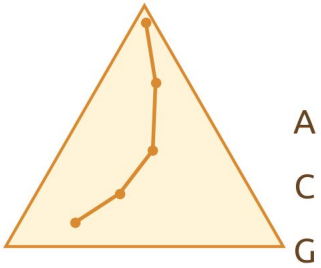
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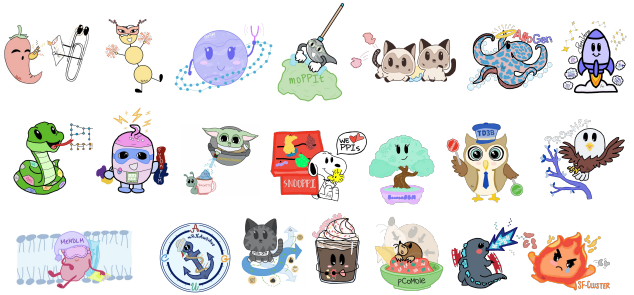
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The master equation, the posterior average, and the rate loss will all return.





And that leads us to
Discrete Flow Matching!

Penn Engineering
University of Pennsylvania

